

Anirudh Maiya

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Summary

Highly skilled Software Engineer and current M.S. in Computer Science candidate at the University of Colorado Boulder with a 4.0 GPA. Proven track record in architecting scalable cloud-native services, optimizing large-scale data processing pipelines, and developing novel machine learning algorithms. Expert in Python, Java, and modern AI frameworks (PyTorch, TensorFlow), with significant experience transitioning complex systems from ideation to production-ready environments. Demonstrated ability to improve system robustness, automate complex workflows, and implement advanced optimization techniques, making me an ideal candidate for driving AI-centric use-cases into production within high-impact enterprise environments.

Work Experience

- Commvault** Boulder, USA
Software Engineer *January 2022 - August 2023*
 - Elevated remote troubleshooting service success rate from 85% to 98% by enhancing software robustness and implementing advanced error handling for PsExec operations.
 - Led a massive migration of a remote troubleshooting service, refactoring 35,000+ lines of code from Python 2.7 to 3.9 and replacing Airflow/Luigi with a high-performance in-house Python codebase.
 - Implemented core software engineering best practices including OOP principles, dependency management, log scrubbing, and generic retry APIs to enhance code adaptability across production and test environments.
 - Served as Subject Matter Expert (SME) for ServiceNow integrations, revamping the Commvault Plugin to meet strict ServiceNow Store regulatory standards.
 - Orchestrated complex library and server migrations, including a Java-based cloud services project upgrade and a manual Tomcat migration involving dependent JAR files.
 - Managed the company's translation editor service, resolving critical user experience issues to boost translator productivity.
- Commvault** Boulder, USA
Associate Software Engineer *July 2021 - December 2021*
 - Architected and deployed 'Decompress-as-a-service' in a cloud ecosystem using a leader-worker architecture, enabling 1,000+ engineers to process 30+ TB of customer logs daily.
 - Implemented on-the-fly maintenance and feature release upgrades for troubleshooting services, saving the Cloud-Services Team approximately 300 manual hours annually.
 - Owned and managed workflows and reporting for the Commvault Cloud platform, serving a user base of over 20,000 customers.
 - Strengthened platform security by decoupling customer-facing reports from licensing code dependencies, mitigating potential vulnerabilities.
- Commvault** Boulder, USA
Development Intern *January 2021 - June 2021*

- Optimized cloud services build performance by migrating dependency management from Apache Ant to Gradle, reducing build times from 3 minutes to 30 seconds.
- Facilitated rapid security response by enabling the organization to promptly address the Log4j vulnerability through improved build agility.
- Designed and automated comprehensive test cases to ensure the robustness of the Commvault Cloud Disaster Recovery solution.

Projects

- **Cloud Image Prior**
Lead Researcher N/A - N/A
 - Devised a novel framework for removing cloud cover from Sentinel-2 remote-sensing imagery without the need for training data.
 - Achieved a significant 64% boost in average Peak Signal-to-Noise Ratio (PSNR).
- **Tom Optimizer**
Developer N/A - N/A
 - Developed a novel variant of the Adam optimizer that accounts for gradient trends.
 - Outperformed traditional optimizers by achieving an average improvement of 4.6% in accuracy and convergence speed.
- **Messup**
Developer N/A - N/A
 - Introduced a novel regularization technique using exponential smoothing for sample encounters.
 - Reduced generalization error by 1.6% when applied to the CIFAR-10 dataset with ResNet architecture.
- **RideShare Application**
Backend Developer N/A - N/A
 - Built a robust backend for ride-sharing, enabling users to create, join, and manage rides.
 - Utilized RabbitMQ for message queuing and orchestrated worker containers for scalable processing.
 - Implemented eventual consistency for database updates and ensured system fault tolerance using Apache ZooKeeper.

Education

- **University of Colorado Boulder** Boulder, USA
Master of Science in Computer Science N/A - May 2025
- **PES University** Boulder, USA
Bachelor of Technology in Computer Science N/A - May 2021

Courses & Certifications

- **Project Intern - Remote Sensing Framework** Indian Space Research Organization (ISRO)
October 2019 - April 2020

Technical Skills

Python, Java, JavaScript, R, SQL, HTML, CSS, PHP, ReactJS, Flask, Node.js, REST API, XML, JSON, Microsoft SQL Server, PostgreSQL, MongoDB, Gradle, RabbitMQ, Apache Spark, Apache ZooKeeper, Git, Linux, PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face, NLTK, Numpy, SpaCy, Plotly, Matplotlib, Azure, Google Cloud Platform, Docker, Kubernetes, Algorithms, Operating Systems

Publications

- Novel Framework for Coconut Farm Area Estimation using
• Multi-spectral Data (Presented at IEEE IGARSS - 2021)

2021

- Cloud Image Prior: Removing cloud cover from Sentinel-2
• remote-sensing without training data

N/A